

## 21.3 Weighted least squares

Once heteroscedasticity has been found, one can transform the response ([Chapter 22](#)), keep least squares and correct its standard errors ([Section 21.4](#)), or give the noisier observations less weight. This section is about weighting. With known variances it is optimal and its theory is already in place. With estimated variances the gain is real but the theory is asymptotic, and a careless implementation can do worse than no weighting at all.

### 21.3.1 Known weights

Suppose  $\text{Var}(\varepsilon_i) = \sigma^2/w_i$ , with the  $w_i > 0$  known and  $\sigma^2$  unknown, and the errors uncorrelated. With  $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$  this is  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{W}^{-1}$ . The generalized least squares estimate of Definition 6.9.1 with  $\mathbf{V} = \mathbf{W}^{-1}$  minimizes

$$\sum_{i=1}^n w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2, \quad \text{so} \quad \hat{\boldsymbol{\beta}}_W = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}. \quad (21.3.1)$$

This is the **weighted least squares** (WLS) estimate. [Section 6.9](#) described its geometry: it is ordinary least squares after each row of  $\mathbf{X}$  and  $\mathbf{y}$  is multiplied by  $\sqrt{w_i}$ , which turns the errors into  $\sqrt{w_i} \varepsilon_i$  with common variance  $\sigma^2$ . Aitken's theorem ([Corollary 7.2.3](#)) makes it the best linear unbiased estimator. Everything in Part III then applies to the rescaled data. The error variance is estimated by

$$s_w^2 = \frac{1}{n-p} \sum_{i=1}^n w_i (y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W)^2,$$

the covariance of  $\hat{\boldsymbol{\beta}}_W$  by  $s_w^2 (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ , and under normality the  $t$  and  $F$  procedures are exact. The residuals to plot are the rescaled ones,  $\sqrt{w_i} (y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W)$ , studentized with the leverages of the weighted fit. The raw residuals  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W$  should still fan out, since their variances still differ.

Known weights arise when  $y_i$  is the mean of  $m_i$  observations ( $w_i = m_i$ ), a total ( $w_i = 1/m_i$ ), or a measurement from an instrument of known precision. The weights are then fixed by the design, not by the data.

### 21.3.2 Estimating the weights

More often the variances are unknown, and the weights must come from the data. There are three common routes.

1. **Replicates.** If there are several observations at each distinct row of  $\mathbf{X}$ , or in each of a few groups known to differ in precision, the sample variance within each group estimates its  $\sigma_g^2$ .
2. **A variance function in other variables.** Model  $\sigma_i^2 = \sigma^2 \exp(\mathbf{z}_i^\top \boldsymbol{\gamma})$ , as in [Eq. 21.2.3](#), and estimate  $\boldsymbol{\gamma}$  by regressing  $\log \hat{\varepsilon}_i^2$  on an intercept and  $\mathbf{z}_i$ . The intercept of that regression is biased, because  $E \log \chi^2(1) \neq 0$ , but weights are needed only up to a common factor, so the intercept does not matter.

3. **A variance function of the mean.** Model  $\sigma_i = \sigma|\mu_i|^\theta$  with  $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , estimate  $\theta$  from the slope of  $\log|\hat{\varepsilon}_i|$  on  $\log|\hat{Y}_i|$ , and weight by  $|\hat{Y}_i|^{-2\theta}$ . Since the weights depend on the fit, the estimate is usually iterated.

The resulting estimate  $\hat{\boldsymbol{\beta}}_{\hat{W}}$  is called **feasible** (or estimated) weighted least squares.

**Example. (Engel's data, reweighted)**

Regressing  $\log \hat{\varepsilon}_i^2$  on an intercept and  $\log \text{income}_i$  gives slope  $\hat{\gamma} = 1.90$ , so the fitted variance is proportional to  $\text{income}^{1.90}$ , close to the square law suggested by Figure 21.2.1. Weighting by  $\text{income}^{-1.90}$  changes the slope from 0.4852 to 0.5711. The model-based standard error  $s_w\{(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}\}_{22}^{1/2}$  of the weighted slope is 0.0149. The sandwich standard error of Section 21.4, which does not trust the variance model, is 0.0172 for the weighted slope and 0.0664 for the unweighted one. So weighting has cut the standard error by a factor of nearly four.

The variance function is not sharply determined by these data. Re-estimating  $\gamma$  from the residuals of the weighted fit, and repeating until it settles (10 iterations), moves  $\hat{\gamma}$  to 2.52 and the slope to 0.5875, about one standard error away. Reporting the weighted estimate with the sandwich standard error protects against this uncertainty in the weights.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])

def wls(X, y, w):
    """Weighted least squares: coefficients, model-based and HC3 covariances."""
    sw = np.sqrt(w)
    Q, R = np.linalg.qr(X * sw[:, None])          # OLS on the whitened data
    b = np.linalg.solve(R, Q.T @ (y * sw))
    ew = (y - X @ b) * sw                          # whitened residuals
    # leverages in the weighted geometry
    hw = np.sum(Q**2, axis=1)
    Rinv = np.linalg.inv(R)
    bread = Rinv @ Rinv.T                          # (X'WX)^{-1}
    s2w = ew @ ew / (len(y) - X.shape[1])
    meat = (Q * (ew / (1 - hw))[:, None]).T @ (Q * (ew / (1 - hw))[:, None])
    return b, s2w * bread, Rinv @ meat @ Rinv.T

b_ols, V_ols, V_ols_hc3 = wls(X, food, np.ones(n))
e = food - X @ b_ols
Z = np.column_stack([np.ones(n), np.log(income)])
# variance ~ income^gamma
gamma = np.linalg.lstsq(Z, np.log(e**2), rcond=None)[0][1]
b_fw, V_fw, V_fw_hc3 = wls(X, food, income**-gamma)
print(f"estimated power gamma = {gamma:.2f}")
print(f"OLS slope {b_ols[1]:.4f} model se {np.sqrt(V_ols[1, 1]):.4f} "
      f"HC3 se {np.sqrt(V_ols_hc3[1, 1]):.4f}")
```

```
print(f"FWLS slope {b_fw[1]:.4f} model se {np.sqrt(V_fw[1, 1]):.4f} "
      f"HC3 se {np.sqrt(V_fw_hc3[1, 1]):.4f}")
```

[Run this code online](#)

### 21.3.3 What feasible weighting achieves

The weights in  $\hat{\beta}_{\hat{W}}$  are random and depend on the same data as the fit, so none of the finite-sample results for known weights carries over directly. The next theorem collects what can be said in general. Part (b) shows that weighting with the wrong weights is harmless for bias. Part (c) is an exact finite-sample result, and part (d) is the large-sample justification in the setting of replicated groups.

#### Theorem 21.3.1 — Weighted least squares with known and estimated weights

Let  $\mathbf{Y} = \mathbf{X}\beta + \epsilon$  with  $E\epsilon = \mathbf{0}$  and  $\text{rank}(\mathbf{X}) = p$ .

- (a) (*Known weights.*) If  $\text{Cov}(\epsilon) = \sigma^2 \mathbf{W}^{-1}$  with  $\mathbf{W}$  known, then  $\hat{\beta}_W$  is the best linear unbiased estimator of  $\beta$ ,  $\text{Cov}(\hat{\beta}_W) = \sigma^2 (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$  and  $E s_w^2 = \sigma^2$ . If  $\epsilon$  is normal,  $\hat{\beta}_W \sim N_p(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1})$  independently of  $(n-p)s_w^2/\sigma^2 \sim \chi^2(n-p)$ .
- (b) (*Wrong weights.*) If  $\text{Cov}(\epsilon) = \Sigma$  is arbitrary and  $\mathbf{W}$  is any fixed positive diagonal matrix, then  $\hat{\beta}_W$  is unbiased with

$$\text{Cov}(\hat{\beta}_W) = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \Sigma \mathbf{W} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}.$$

- (c) (*Estimated weights, symmetric errors.*) Let  $\hat{\mathbf{W}} = \mathbf{W}(\hat{\epsilon})$  be a positive diagonal matrix that depends on the data only through the ordinary least squares residuals, with  $\mathbf{W}(-\epsilon) = \mathbf{W}(\epsilon)$ . If  $-\epsilon$  has the same distribution as  $\epsilon$  and  $E\|\hat{\beta}_{\hat{W}}\| < \infty$ , then  $E\hat{\beta}_{\hat{W}} = \beta$ .
- (d) (*Estimated weights from groups.*) Suppose the observations form  $G$  groups, fixed in advance, that the errors are independent, and that within group  $g$  they are identically distributed with variance  $\sigma_g^2 > 0$ . Let  $n_g \rightarrow \infty$  for each  $g$  with  $n^{-1} \mathbf{X}_g^\top \mathbf{X}_g \rightarrow \mathbf{A}_g$ , where  $\mathbf{X}_g$  holds the rows of group  $g$ , and let  $\mathbf{A}_w = \sum_g \mathbf{A}_g / \sigma_g^2$  be positive definite. Let  $\mathbf{W}$  have entries  $1/\sigma_g^2$ , and  $\hat{\mathbf{W}}$  have entries  $1/\hat{\sigma}_g^2$ , where  $\hat{\sigma}_g^2 = \|\hat{\epsilon}_g\|^2/n_g$  is the mean squared ordinary least squares residual in group  $g$ . Then

$$\sqrt{n}(\hat{\beta}_{\hat{W}} - \hat{\beta}_W) \rightarrow \mathbf{0} \quad \text{and} \quad n(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \rightarrow \mathbf{A}_w^{-1}$$

in probability, while  $n \text{Cov}(\hat{\beta}_W) = n(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \rightarrow \mathbf{A}_w^{-1}$ .

*Proof.*

- (a) Multiply Eq. 21.3.1 through by  $\mathbf{W}^{1/2}$ : the model  $\mathbf{W}^{1/2} \mathbf{Y} = \mathbf{W}^{1/2} \mathbf{X} \beta + \mathbf{W}^{1/2} \epsilon$  satisfies  $\text{Cov}(\mathbf{W}^{1/2} \epsilon) = \sigma^2 \mathbf{I}$ , and  $\hat{\beta}_W$  is its least squares estimate. The claims are Corollary 7.2.3, Theorem 5.5.1, Theorem 5.6.1 and Theorem 7.5.1 applied to the rescaled model.
- (b)  $\hat{\beta}_W - \beta = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \epsilon$  is a fixed linear map of  $\epsilon$ ; apply Theorem 2.2.1.

- (c) The residuals  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$  do not involve  $\boldsymbol{\beta}$ , and they change sign with  $\boldsymbol{\varepsilon}$ . So  $\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}} - \boldsymbol{\beta} = \mathbf{g}(\boldsymbol{\varepsilon})$  with  $\mathbf{g}(\boldsymbol{\varepsilon}) = (\mathbf{X}^\top \mathbf{W}(\hat{\boldsymbol{\varepsilon}}) \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}(\hat{\boldsymbol{\varepsilon}}) \boldsymbol{\varepsilon}$ , and  $\mathbf{g}(-\boldsymbol{\varepsilon}) = -\mathbf{g}(\boldsymbol{\varepsilon})$  because  $\mathbf{W}(-\hat{\boldsymbol{\varepsilon}}) = \mathbf{W}(\hat{\boldsymbol{\varepsilon}})$ . If  $-\boldsymbol{\varepsilon}$  has the law of  $\boldsymbol{\varepsilon}$ , then  $\mathbf{g}(\boldsymbol{\varepsilon})$  has the law of  $\mathbf{g}(-\boldsymbol{\varepsilon}) = -\mathbf{g}(\boldsymbol{\varepsilon})$ , and its mean, which exists, equals its negative.
- (d) *Consistency of the group variances.* Write  $\mathbf{a} = \mathbf{M}\boldsymbol{\varepsilon}$ , so  $\hat{\boldsymbol{\varepsilon}}_g = \boldsymbol{\varepsilon}_g - \mathbf{a}_g$ . By the weak law,  $\|\boldsymbol{\varepsilon}_g\|^2/n_g \rightarrow \sigma_g^2$ . Also  $E\|\mathbf{a}\|^2 = \text{tr}(\mathbf{M}\boldsymbol{\Sigma}) \leq p \max_g \sigma_g^2$ , so  $\|\mathbf{a}_g\|^2/n_g \leq \|\mathbf{a}\|^2/n_g \rightarrow 0$  in probability. The triangle inequality  $|\|\hat{\boldsymbol{\varepsilon}}_g\| - \|\boldsymbol{\varepsilon}_g\|| \leq \|\mathbf{a}_g\|$  gives  $\hat{\sigma}_g^2 \rightarrow \sigma_g^2$ , hence  $\hat{w}_g = 1/\hat{\sigma}_g^2 \rightarrow w_g = 1/\sigma_g^2$ .

*Comparison.* Put  $\mathbf{A}_n(\mathbf{v}) = n^{-1} \sum_g v_g \mathbf{X}_g^\top \mathbf{X}_g$  for weights  $\mathbf{v} = (v_1, \dots, v_G)$ , and  $\mathbf{b}_g = n^{-1/2} \mathbf{X}_g^\top \boldsymbol{\varepsilon}_g$ . Then

$$\begin{aligned} \sqrt{n}(\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}} - \boldsymbol{\beta}) &= \mathbf{A}_n(\hat{\mathbf{w}})^{-1} \sum_g \hat{w}_g \mathbf{b}_g, \\ \sqrt{n}(\hat{\boldsymbol{\beta}}_{\mathbf{W}} - \boldsymbol{\beta}) &= \mathbf{A}_n(\mathbf{w})^{-1} \sum_g w_g \mathbf{b}_g. \end{aligned}$$

Each  $\mathbf{b}_g$  has mean zero and  $E\|\mathbf{b}_g\|^2 = \sigma_g^2 \text{tr}(n^{-1} \mathbf{X}_g^\top \mathbf{X}_g)$ , which is bounded, so  $\mathbf{b}_g = O_p(1)$ . Both  $\mathbf{A}_n(\hat{\mathbf{w}})$  and  $\mathbf{A}_n(\mathbf{w})$  converge in probability to  $\mathbf{A}_w$ , which is invertible, and matrix inversion is continuous there. The difference of the two displays is

$$\{\mathbf{A}_n(\hat{\mathbf{w}})^{-1} - \mathbf{A}_n(\mathbf{w})^{-1}\} \sum_g \hat{w}_g \mathbf{b}_g + \mathbf{A}_n(\mathbf{w})^{-1} \sum_g (\hat{w}_g - w_g) \mathbf{b}_g,$$

a sum of products of  $o_p(1)$  and  $O_p(1)$  terms, so it is  $o_p(1)$ . Finally  $n(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} = \mathbf{A}_n(\hat{\mathbf{w}})^{-1} \rightarrow \mathbf{A}_w^{-1}$ , and  $n(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} = \mathbf{A}_n(\mathbf{w})^{-1} \rightarrow \mathbf{A}_w^{-1}$ . ■

Part (c) is exact but covers only weights computed from the least squares residuals, not those computed from fitted values ([Exercise \(B3\)](#)). Part (d) says that with enough replicates estimating the weights costs nothing in large samples. The scaled errors  $\sqrt{n}(\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}} - \boldsymbol{\beta})$  and  $\sqrt{n}(\hat{\boldsymbol{\beta}}_{\mathbf{W}} - \boldsymbol{\beta})$  differ by  $o_p(1)$ , so whatever limiting distribution the optimal estimator has, the feasible one has the same; it is normal when, for instance, the rows of  $\mathbf{X}$  are bounded, by Lemma 21.2.1(a) applied within each group. The model-based covariance estimate is consistent as well. Carroll and Ruppert (1988) prove the same conclusion for smooth parametric variance functions, including functions of the mean, under conditions we do not reproduce. What (d) does not say is how many replicates are “enough”. A simulation shows that the answer can be surprisingly many.

#### Example. (Weights from a few replicates)

Take ten design points  $x = 1, \dots, 10$ ,  $m$  replicates at each, a straight-line mean, and normal errors with standard deviation proportional to  $x$ . [Figure 21.3.1](#) compares four slope estimates by their variance relative to weighted least squares with the true weights. Ordinary least squares has 2.44 times the optimal variance when  $m = 2$ , and about the same for every  $m$ . Weights  $1/s_g^2$  from the within-group sample variances are a disaster for small groups: with  $m = 2$  their variance is 5.44 times the optimum, twice that of the unweighted fit, and with  $m = 3$  it is still 2.67, worse than no weighting. They approach the optimum only slowly (1.22 for  $m = 10$ , 1.05 for  $m = 30$ ). Worse, the model-based interval, which treats the estimated weights as known, covers the true slope in only 0.47 of the data sets for  $m = 2$ , 0.65 for  $m = 3$  and 0.80 for  $m = 5$ . Weights from a fitted variance function, the regression of  $\log s_g^2$  on  $\log x_g$ , are far better: 1.41 times the

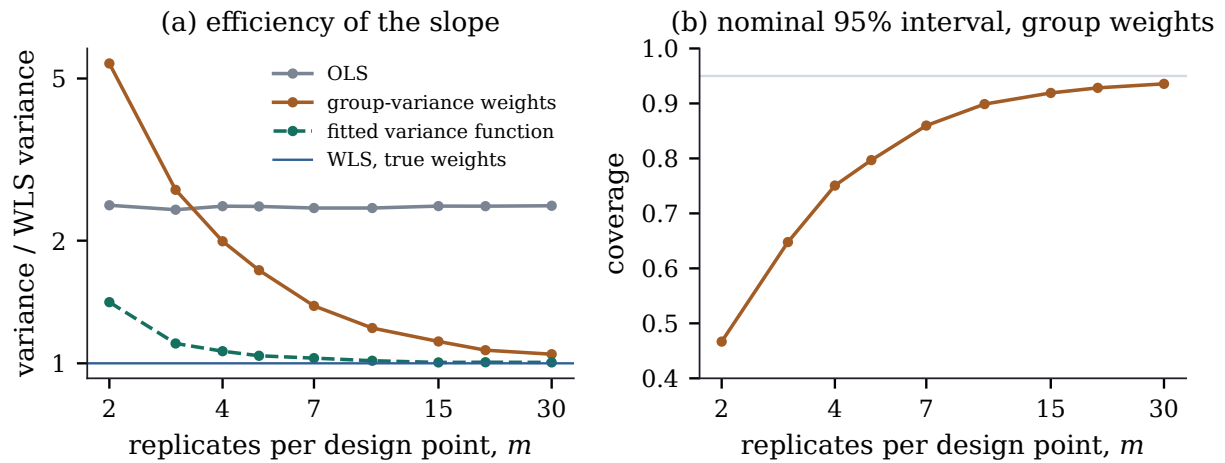


Figure 21.3.1: Feasible weighted least squares with  $m$  replicates at each of ten design points and standard deviation proportional to  $x$  (20,000 simulated data sets for each  $m$ ). (a) Variance of the slope estimate relative to weighted least squares with the true weights. (b) Coverage of the nominal 95% interval computed as if the within-group weights were known.

optimum for  $m = 2$ , 1.12 for  $m = 3$  and 1.04 for  $m = 5$ .

```
rng = np.random.default_rng(2106)
# ten design points, sd proportional to x
xg = np.arange(1.0, 11.0)

def replicate_study(m, reps):
    """Variance of four slope estimators relative to WLS with true weights."""
    x = np.repeat(xg, m)
    Xr = np.column_stack([np.ones(len(x)), x])
    Y = 1 + 0.5 * x + rng.normal(size=(reps, len(x))) * x

    def slopes(W):
        A = np.einsum("ri,ij,ik->rjk", W, Xr, Xr) # X'WX for each replicate
        c = np.einsum("ri,ij,ri->rj", W, Xr, Y)
        return np.linalg.solve(A, c[:, None])[..., 0], A

    b_o, _ = slopes(np.ones_like(Y))
    b_w, _ = slopes(np.tile(1 / x**2, (reps, 1)))
    # within-group variances
    s2g = Y.reshape(reps, len(xg), m).var(axis=2, ddof=1)
    Wg = np.repeat(1 / s2g, m, axis=1)
    b_g, A_g = slopes(Wg)
    # smooth model: log s2 on log x
    Zg = np.column_stack([np.ones(len(xg)), np.log(xg)])
    coef = np.linalg.lstsq(Zg, np.log(s2g).T, rcond=None)[0]
    b_m, _ = slopes(np.repeat(np.exp(-(Zg @ coef).T), m, axis=1))
    var = np.array([np.var(b[:, 1]) for b in (b_o, b_w, b_g, b_m)])
    # nominal interval for b_g
    res = Y - b_g @ Xr.T
    se = np.sqrt((Wg * res**2).sum(axis=1) / (len(x) - 2)
                 * np.linalg.inv(A_g)[:, 1, 1])
    q = stats.t.ppf(0.975, len(x) - 2)
    cover = np.mean(np.abs(b_g[:, 1] - 0.5) <= q * se)
    return var / var[1], cover
```

```

for m in (2, 3, 5, 10, 30):
    rel, cover = replicate_study(m, 2000)
    print(f"m = {m:2d}: variance relative to WLS  OLS {rel[0]:.2f} "
          f"group weights {rel[2]:.2f}  fitted weights {rel[3]:.2f}; "
          f"coverage {cover:.2f}")

```

[Run this code online](#)

The failure of the group weights has a simple cause. With  $m = 2$  each  $s_g^2$  is  $\sigma_g^2 \chi^2(1)$ , which is often tiny, and  $1/s_g^2$  then gives one group an enormous weight on the strength of two nearly equal responses. In fact  $E(1/s_g^2) = \infty$  ([Exercise \(C1\)](#)). A fitted variance function pools the information from all groups into a few parameters and cannot be fooled by one lucky pair. The general lessons are to estimate a smooth variance function whenever possible, and to report feasible weighted estimates with standard errors that do not depend on the variance model being right. Part (b) of the theorem shows what those are: the sandwich form with  $\Sigma$  estimated from the residuals, which is the subject of the next section.

## 21.3.4 Exercises

### 21.3.4.1 A. Check your understanding

#### Exercise A1.

Under [Theorem 21.3.1\(a\)](#), show that  $E \sum_i w_i (Y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W)^2 = \sigma^2(n - p)$ , and that the raw residual  $Y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W$  has variance  $\sigma^2(1 - g_{ii})/w_i$ , where  $g_{ii}$  is the  $i$ th leverage of the weighted fit.

#### Exercise A2.

A laboratory reports only the mean  $\bar{y}_i$  of  $m_i$  replicate measurements at each of  $k$  settings  $\mathbf{x}_{(i)}$ , where the individual measurements have common variance  $\sigma^2$ . Which weights should be used? What does  $s_w^2$  then estimate, and why is it not the replicate variance  $\sigma^2$  unless the model for the means is correct?

### 21.3.4.2 B. Practice

#### Exercise B1.

In the model  $y_i = \beta x_i + \varepsilon_i$  with  $x_i > 0$  and  $\text{Var}(\varepsilon_i) = \sigma^2 x_i^2$ , show that the weighted estimate is the mean of the ratios  $y_i/x_i$ , and that the efficiency of ordinary least squares relative to it is  $(\sum_i x_i^2)^2 / (n \sum_i x_i^4) \leq 1$ . Evaluate the efficiency for  $x = 1, \dots, 10$ .

*Solution.* With  $w_i = x_i^{-2}$ ,  $\hat{\beta}_W = \sum_i w_i x_i y_i / \sum_i w_i x_i^2 = n^{-1} \sum_i y_i / x_i$ , with variance  $\sigma^2/n$ . Ordinary least squares gives  $\sum_i x_i y_i / \sum_i x_i^2$ , with variance  $\sigma^2 \sum_i x_i^4 / (\sum_i x_i^2)^2$ . The ratio is  $(\sum_i x_i^2)^2 / (n \sum_i x_i^4)$ , at most one by the Cauchy–Schwarz inequality applied

to the vectors  $(x_i^2)$  and  $(1)$ . For  $x = 1, \dots, 10$ ,  $\sum x_i^2 = 385$  and  $\sum x_i^4 = 25,333$ , so the efficiency is  $148,225/253,330 = 0.585$ .

### Exercise B2.

Some programs implement integer weights  $w_i$  by repeating row  $i$  of the data  $w_i$  times and running ordinary least squares. Show that this gives the correct  $\hat{\beta}_W$  but divides the weighted residual sum of squares by  $\sum_i w_i - p$  instead of  $n - p$ . Which standard errors are then wrong, and in which direction?

### Exercise B3.

Weights  $\hat{w}_i = |\hat{Y}_i|^{-2\theta}$  depend on the fitted values. Show that they are not functions of  $\hat{\epsilon}$  alone, and find where the symmetry argument of [Theorem 21.3.1\(c\)](#) fails. What happens for a single regressor through the origin,  $Y_i = \beta x_i + \epsilon_i$  with every  $x_i \neq 0$ ?

*Solution.*  $\hat{Y} = \mathbf{X}\beta + \mathbf{M}\epsilon$  depends on  $\beta$ , and replacing  $\epsilon$  by  $-\epsilon$  turns it into  $\mathbf{X}\beta - \mathbf{M}\epsilon$ , which does not have the same absolute entries. So  $\hat{W}$  is not an even function of  $\epsilon$ , and  $\mathbf{g}(-\epsilon) \neq -\mathbf{g}(\epsilon)$  in general. The feasible estimate then need not be unbiased in finite samples, and the argument of [Theorem 21.3.1\(c\)](#) gives no guarantee.

The single regressor through the origin is an exception. There  $\hat{Y}_i = \hat{\beta}x_i$ , so  $\hat{w}_i = |\hat{\beta}|^{-2\theta}|x_i|^{-2\theta}$  whenever  $\hat{\beta} \neq 0$ , an event of probability one for continuously distributed errors. The random factor  $|\hat{\beta}|^{-2\theta}$  is common to all the weights and cancels from  $\hat{\beta}_W = \sum_i \hat{w}_i x_i Y_i / \sum_i \hat{w}_i x_i^2$ . So the feasible estimate equals weighted least squares with the fixed weights  $|x_i|^{-2\theta}$ . It is unbiased by [Theorem 21.3.1\(b\)](#), and if  $\text{Var}(\epsilon_i)$  is proportional to  $|x_i|^{2\theta}$  it is the best linear unbiased estimator by [Theorem 21.3.1\(a\)](#); for  $\theta = 1$  this is the ratio-type case of variance proportional to  $x_i^2$ . With two or more regressors the ratios  $\hat{w}_i/\hat{w}_j = |\hat{Y}_j/\hat{Y}_i|^{2\theta}$  depend on the data, and the argument fails again.

#### 21.3.4.3 C. Going deeper

### Exercise C1.

Show that  $E(1/U) = \infty$  if  $U \sim \chi^2(1)$ , and that  $E(1/U) = 1/(k-2)$  if  $U \sim \chi^2(k)$  with  $k \geq 3$ . Explain what this implies for weights  $1/s_g^2$  computed from  $m$  replicates.

*Solution.* The density of  $\chi^2(k)$  is proportional to  $u^{k/2-1}e^{-u/2}$ , so  $E(1/U)$  is proportional to  $\int_0^\infty u^{k/2-2}e^{-u/2} du$ , which diverges at zero when  $k/2 - 2 \leq -1$ , that is, for  $k \leq 2$ . For  $k \geq 3$  the integral is a gamma integral and gives  $E(1/U) = \Gamma(k/2 - 1)/\{2\Gamma(k/2)\} = 1/(k-2)$ . With  $m$  normal replicates,  $(m-1)s_g^2/\sigma_g^2 \sim \chi^2(m-1)$ , so  $1/s_g^2$  has infinite mean for  $m \leq 3$  and mean  $(m-1)/\{(m-3)\sigma_g^2\}$  for  $m \geq 4$ . The group weights are both inflated and wildly variable in small groups.